

ABSTRACT

This thesis develops SafeSAC, a physics-aware safe deep reinforcement learning (DRL) controller that coordinates bidirectional electric-vehicle (EV) charging for voltage support on weak distribution feeders. On a modified IEEE 33-bus Baran–Wu feeder it matches the safety of a strong Lagrangian baseline while roughly doubling the EV state-of-charge service it delivers, and it remains operational when deployed on a grid different from the one it was trained on.

Mass EV adoption concentrates large, controllable loads on radial feeders that were never designed for them. On weak feeders—long, high resistance-to-reactance (R/X) lines with low short-circuit capacity—uncoordinated charging drives bus voltages outside the statutory $[0.95, 1.05]$ pu band, while vehicle-to-grid (V2G) discharge is an untapped lever for local voltage support. We cast real-time charging as a constrained Markov decision process (CMDP) over a 288-step day, with a reward combining time-of-use energy cost, V2G revenue, and battery degradation, and a per-step cost that measures voltage-band and transformer violations. SafeSAC couples an augmented-Lagrangian Soft Actor-Critic (SAC-Lag) agent with a safety-projection layer that solves an eight-variable second-order cone program (SOCP) at every control step. The projection is built from power-flow voltage sensitivities measured on the actual operating network, so it adapts to the deployment grid’s R/X structure instead of assuming a strong grid. The controller is benchmarked against an uncoordinated heuristic, an IEEE 1547-2018 droop controller, and the SAC-Lag baseline; all learned agents are trained at load scale 0.50 and evaluated on 25 shared-seed episodes for a paired comparison.

On the weak feeder the projection is demonstrably grid-aware: at the most sensitive station it curtails a -80 kW V2G request to -54 kW, whereas on a strong grid the same request passes essentially unaltered. Relative to SAC-Lag, SafeSAC holds the in-distribution voltage-violation rate statistically constant (0.091 vs. 0.090; paired t -test $p = 0.86$) while raising the fraction of EV state-of-charge targets met from 0.277 to 0.569 (+29 percentage points; Cohen’s $d = 2.04$; $p \ll 0.001$). Under a strong-to-weak deployment shift the SAC-Lag policy collapses to a non-charging, drain-and-sell behaviour (targets met = 0.000), whereas SafeSAC stays operational and delivers +45 percentage points more service. A pre-registered protocol of five decision gates returns three passes, and the two non-passing gates are shown to be single-axis tests whose outcomes in fact strengthen the case for the method.

The results indicate that embedding lightweight, network-specific physics into the action pathway of a DRL agent yields controllers that are simultaneously safe and useful on the weak rural and peri-urban feeders where EV coordination matters most, supporting renewable integration and deferring costly network reinforcement.

Keywords: safe reinforcement learning; soft actor-critic; vehicle-to-grid; voltage support; weak distribution feeder; second-order cone program; physics-aware control; constrained Markov decision process.